

TRANSPLANTATION

MAIT and V δ 2 unconventional T cells are supported by a diverse intestinal microbiome and correlate with favorable patient outcome after allogeneic HCT

Hana Andrlóv¹, Oriana Miltiadous^{2†}, Anastasia I. Kousa^{1†}, Anqi Dai¹, Susan DeWolf³, Sara Violante⁴, Hee-Yon Park⁴, Sudha Janaki-Raman⁴, Rui Gardner¹, Sary El Daker⁵, John Slingerland¹, Paul Giardina¹, Annelie Clurman¹, Antonio L. C. Gomes¹, Chi Nguyen^{1,6}, Marina Burgos da Silva¹, Gabriel K. Armijo¹, Nicole Lee¹, Roberta Zappasodi^{7,8}, Ronan Chaligne⁹, Ignas Masilionis⁹, Emily Fontana¹, Doris Ponce^{8,10}, Christina Cho^{8,10}, Amy Bush¹¹, Lauren Hill¹¹, Nelson Chao¹¹, Anthony D. Sung¹¹, Sergio Giralt^{8,10}, Esther H. Vidal², Kinga K. Hosszu², Sean M. Devlin¹², Jonathan U. Peled^{8,10}, Justin R. Cross⁴, Miguel-Angel Perales^{8,10}, Dale I. Godfrey¹³, Marcel R. M. van den Brink^{1,8,10†}, Kate A. Markey^{8,10,14,15*}†

Microbial diversity is associated with improved outcomes in recipients of allogeneic hematopoietic cell transplantation (allo-HCT), but the mechanism underlying this observation is unclear. In a cohort of 174 patients who underwent allo-HCT, we demonstrate that a diverse intestinal microbiome early after allo-HCT is associated with an increased number of innate-like mucosal-associated invariant T (MAIT) cells, which are in turn associated with improved overall survival and less acute graft-versus-host disease (aGVHD). Immune profiling of conventional and unconventional immune cell subsets revealed that the prevalence of V δ 2 cells, the major circulating subpopulation of $\gamma\delta$ T cells, closely correlated with the frequency of MAIT cells and was associated with less aGVHD. Analysis of these populations using both single-cell transcriptomics and flow cytometry suggested a shift toward activated phenotypes and a gain of cytotoxic and effector functions after transplantation. A diverse intestinal microbiome with the capacity to produce activating ligands for MAIT and V δ 2 cells appeared to be necessary for the maintenance of these populations after allo-HCT. These data suggest an immunological link between intestinal microbial diversity, microbe-derived ligands, and maintenance of unconventional T cells.

INTRODUCTION

Unconventional T cells are potent early responder T cells that act as a bridge between innate and adaptive immunity (1). These populations are equipped with a T cell receptor (TCR) like conventional T cells, but they do not recognize peptide antigens presented in the context of highly polymorphic major histocompatibility complex (MHC) molecules and are instead activated by lipid and metabolite antigens presented by conserved antigen-presenting molecules (1). These antigens typically go undetected by the conventional T cell subtypes. Mucosal-associated invariant T (MAIT) cells recognize metabolites derived from the microbial riboflavin pathway presented as antigens in the context of the ubiquitously expressed MHC-class I-related protein MR1 (2). Natural killer T (NKT) cells recognize microbial and self-glycolipids presented by CD1d (3), and V δ 2 cells, the major circulating subset of human $\gamma\delta$ T cells, respond to intermediates in the 2-C-methyl-D-erythritol 4-phosphate (MEP) isoprenoid

synthesis pathway (4). Of these, the molecule 4-hydroxy-3-methylbut-2-enyl pyrophosphate (HMBPP) is the most potent activating ligand and binds to the intracellular domain of butyrophilin 3A1 and 2A1, which then undergoes a confirmation change that is recognized by the $\gamma\delta$ TCR (5, 6). The antimicrobial role of unconventional T cells in the context of infection is well recognized, however, the role of the commensal microbiota in regulating unconventional T cell homeostasis is less well understood (7–10).

Studies from our group and others (11–13) have associated features of the early posttransplantation intestinal microbiome with overall survival and complications related to allogeneic hematopoietic stem cell transplantation (allo-HCT). Allo-HCT is a curative-intent therapy for hematological cancers that involves immunoablative pretreatment of the recipient with chemotherapy and/or radiation followed by infusion of an allogeneic hematopoietic stem cell graft from either a family member or unrelated donor. Despite advances in the field,

¹Department of Immunology, Sloan Kettering Institute, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ²Department of Pediatrics, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ³Leukemia Service, Department of Medicine, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ⁴Donald B. and Catherine C. Marron Cancer Metabolism Center, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ⁵Department of Pathology, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ⁶Louis V. Gerstner Jr. Graduate School of Biomedical Sciences, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ⁷Human Oncology Pathogenesis Program, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ⁸Weill Cornell Medical College, New York, NY, USA. ⁹Program for Computational and Systems Biology, Sloan Kettering Institute, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ¹⁰Adult Bone Marrow Transplantation Service, Department of Medicine, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ¹¹Division of Hematologic Malignancies and Cellular Therapy, Department of Medicine, Duke University Medical Center, Durham, NC, USA. ¹²Department of Epidemiology and Biostatistics, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ¹³Department of Microbiology and Immunology, The Peter Doherty Institute for Infection and Immunity, University of Melbourne, Melbourne, Victoria, Australia. ¹⁴Clinical Research Division, Fred Hutchinson Cancer Research Center (FHCRC), Seattle, WA, USA. ¹⁵Division of Medical Oncology, University of Washington, Seattle, WA, USA.

*Corresponding author: Email: kmarkey@fredhutch.org

†These authors contributed equally to this work.

the broader implementation of allo-HCT is limited by high rates of transplantation-related complications, with 3-year overall mortality approaching 50% (14). Intestinal microbial α -diversity is a commonly used summary metric reflecting both the number of different taxa present and their distribution within a sample (15). A diverse intestinal microbiome, measured by a higher α -diversity, during the peri-engraftment period (days 7 to 21 after cell infusion) has been associated with favorable transplantation outcomes (11). These include lower risks of infection, relapse of the underlying malignancy, and incidence of graft-versus-host disease (GVHD), an immune-mediated condition in which the donor T cells attack host tissues (11, 12, 16).

Unconventional T cells have restricted TCRs and lack the donor-MHC restriction that is characteristic of the reconstituting conventional donor T cell subsets with undesirable alloreactive potential. They also have been reported to have antitumor effects and have the capacity for antigen-independent activation (i.e., via cytokine signaling) (1). With the exception of NKT cells, which have been well studied in mouse models of allo-HCT (17–19), a key challenge of studying unconventional T cells is that mouse models do not faithfully mimic the biology of these cell types in humans. Recipient MAIT cells have been studied in mouse models of GVHD and have been shown to have a protective effect (20). However, the number of circulating MAIT cells is markedly lower in mice than in humans, and characterization of their function has been generally limited to tissue-resident cells. Differences in the $\gamma\delta$ subsets are even more notable between mice and humans. The $V\delta 2^+$ cells (which are almost exclusively $V\gamma 9^+$) are the most prevalent circulating $\gamma\delta$ T cell subset in humans but lack a mouse homolog (21), making functional studies of this cell type possible only by using xenograft mouse models and humanized mice (22–24). For these reasons, characterization of primary human samples is crucial to deciphering the contributions of these unconventional T cell populations to immune function (25). In humans, NKT cell numbers have been associated with lower rates of GVHD (26, 27). $\gamma\delta$ T cells confer a strong antitumor and anti-infectious potential, without aggravating or even protecting against GVHD (28–32). High MAIT cell counts in infused grafts and post-allo-HCT peripheral blood samples have been associated with lower rates of acute and chronic GVHD, although this has not been a consistent finding in the studies published thus far (33–38). To date, only three human studies in the allo-HCT setting have been able to link MAIT cell reconstitution with the intestinal microbiota, each with a limited number of stool samples (36–38).

Using multiparameter flow cytometry on peripheral blood samples and stool microbiome analysis, we demonstrate an association between specific early microbiome features and early MAIT cell frequencies after allo-HCT. Higher early MAIT cell frequency is associated with less acute GVHD (aGVHD), decreased transplantation-related mortality, and prolonged overall survival. We describe that the $V\delta 2$ subset of $\gamma\delta$ T cells correlates with MAIT cells in the posttransplantation setting, and both appear dependent on intestinal microbial features and microbiome-derived ligands. These findings suggest that strategies to support microbial diversity in the early posttransplantation period are likely to improve maintenance and recovery of protective unconventional T cell populations, which may be one of the mechanisms by which a diverse microbiome supports favorable long-term patient outcomes after allo-HCT.

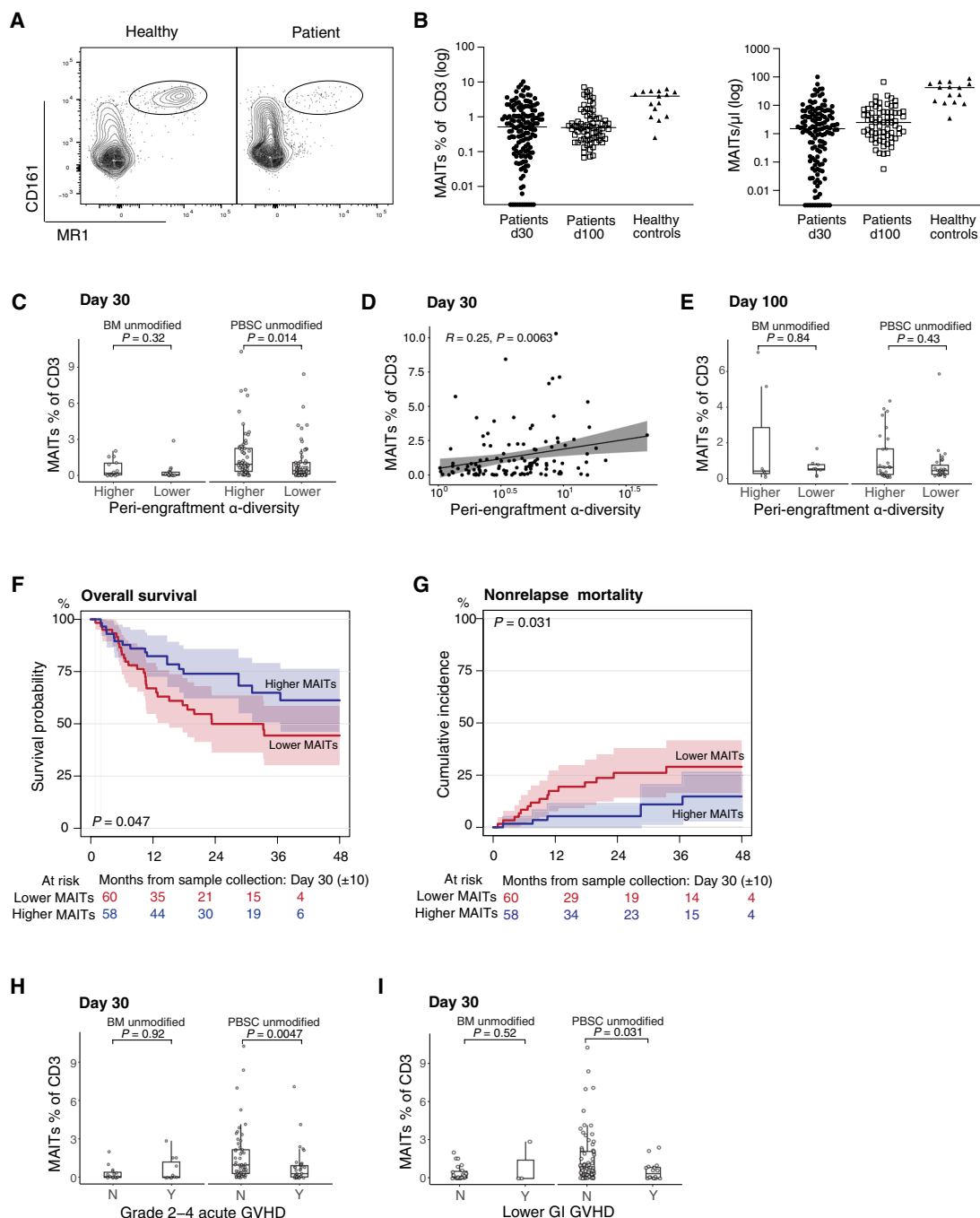
RESULTS
Frequency of MAIT cells early after transplantation is associated with diverse intestinal microbiome and correlates with favorable transplantation outcome

We assembled a cohort of patients ($n = 174$) undergoing allo-HCT at our center and performed multiparameter immunophenotyping and stool microbial profiling. The patient cohort is described in fig. S1, and patient characteristics are described in Table 1. Early MAIT cell reconstitution has previously been associated with favorable outcome after HCT, so we first analyzed the MAIT cell frequency within the $CD3^+$ T cell compartment and their absolute number (cells per microliter) at days 30 and 100 after transplantation by staining with the MR1 tetramer in combination with CD161 (Fig. 1, A and B; complete gating strategy shown in fig. S2), capturing predominantly mature MAIT cells ($CD45RO^+CCR7^-$). We also identified MAIT cells using an antibody against the invariant $V\alpha 7.2$ TCR and CD161, which is an alternative staining strategy that is less specific but commonly used, and this analysis produced similar results (fig. S3, A to D). A number of patients (45 patients, 23.4% of the cohort) had samples at both time points and, consistent with others (36), we observed that there

Table 1. Patient characteristics.		
	All patients	PBSC with d30 sample
	<i>n</i> = 174	<i>n</i> = 118
Sex = M (%)	112 (64.4)	75 (63.6)
Age [mean (SD)]	58.6 (12.84)	60.27 (11.22)
Disease (%)		
Leukemia	77 (44.3)	57 (48.3)
MDS/MPN	42 (24.1)	29 (24.6)
Lymphoma	45 (25.9)	30 (25.4)
Multiple myeloma	4 (2.3)	2 (1.7)
Nonmalignant hematological disorders	6 (3.4)	
Conditioning intensity (%)		
Reduced intensity	89 (51.1)	72 (61.0)
Ablative	56 (32.2)	28 (23.7)
Nonablative	29 (16.7)	18 (15.3)
Donor HLA match (%)		
Unrelated identical	88 (50.6)	65 (55.1)
Related identical sibling	45 (25.9)	35 (29.7)
Related haploidentical	27 (15.5)	7 (5.9)
Unrelated nonidentical	14 (8.0)	11 (9.3)
Graft source		
PBSC unmodified (%)	139 (79.9)	118 (100.0)
BM unmodified (%)	35 (20.1)	
GVHD = Y (%)	104 (59.8)	71 (60.2)
Relapse = Y (%)	47 (27.0)	33 (28.0)

Fig. 1. MAIT cells are supported by a diverse intestinal microbiome after HCT and predict favorable patient outcomes.

(A) FACS plots of MAIT cells (MR1⁺/CD161⁺) in a healthy volunteer and a representative allo-HCT patient. (B) MAIT cell frequencies among CD3⁺ cells and absolute counts on day 30 (± 10) and day 100 (± 20) after HCT (frequencies: day 30 $n = 147$, day 100 $n = 72$; healthy $n = 15$; absolute counts: day 30 $n = 137$, day 100 $n = 70$, healthy $n = 15$). Each symbol represents a single patient sample, lines represent median. (C) Recipients of BM or PBSC grafts were classified into $>$ median (higher) or $\leq$ median (lower) α -diversity and MAIT frequency at day 30 was compared (Wilcoxon rank-sum test, BM $n = 15 \leq$ median, $n = 14 >$ median, $P = 0.32$; PBSC $n = 59 \leq$ median, $n = 59 >$ median, $P = 0.014$). (D) Stool α -diversity at d7 to d21 as a continuous variable correlated with MAIT frequency at day 30 (Pearson method; $n = 118$, $R = 0.25$, $P = 0.0063$). Each dot represents an individual patient sample. (E) Patients were divided as in (C), data shown for day 100 MAIT frequency (Wilcoxon rank-sum test, BM, $n = 8 \leq$ median α -diversity, $n = 8 >$ median, $P = 0.84$; PBSC, $n = 28 \leq$ median α -diversity, $n = 28 >$ median, $P = 0.43$). (F) Kaplan-Meier survival analysis of PBSC-graft recipients stratified by MAIT cell frequency (log-rank test, $n = 118$, $P = 0.047$). (G) Cumulative incidence of NRM in PBSC-graft recipients with $>$ median or $\leq$ median MAIT cell frequency (Gray's test, $n = 118$, $P = 0.031$). (H) Recipients of either BM or PBSC grafts were classified by GVHD status (grades 2 to 4, Y = Yes, grades 2 to 4 GVHD present, N = No, no grades 2 to 4 GVHD). MAIT cell frequency at day 30 shown (Wilcoxon rank-sum test, BM, $n = 15$ grades 0 and 1 GVHD, $n = 11$, grades 2 to 4, $P = 0.92$; PBSC, $n = 53$ grades 0 and 1 GVHD, $n = 36$ with grades 2 to 4, $P = 0.0047$). (I) BM or PBSC recipients were classified by the presence of lower GI GVHD (Y = Yes, lower GI GVHD present or N = No, no lower GI GVHD); MAIT cell frequencies are shown (BM, $n = 23$ without lower GI GVHD, $n = 3$ with lower GI GVHD, $P = 0.52$; PBSC, $n = 73$ without lower GI GVHD, $n = 16$ with any stage lower GI GVHD, $P = 0.031$). (H and I) Patients with blood samples collected before or on the day of GVHD onset were included. (C, E, H and I) Each dot represents a single patient, and boxes represent median with interquartile range.



was no significant increase in MAIT frequency and absolute numbers between days 30 and 100 after transplantation (fig. S4, A and B). We analyzed the data as MAIT cell frequency of CD3⁺ cells rather than absolute cell counts, consistent with previous reports for other unconventional subsets (26). This frequency-based analysis is consistent with data derived from numerical counts, as shown in fig. S5, A to D.

We next asked whether above-median peri-engraftment fecal bacterial diversity was associated with MAIT cell frequency at days 30 and 100 after transplantation, and we compared patients who received unmodified bone marrow (BM) grafts versus recipients of peripheral blood stem cell (PBSC) grafts. MAIT cells were approximately twofold less frequent on day 30 in the samples from recipients of BM grafts compared with PBSC, a difference that was no longer

evident on day 100 (fig. S6, A and B, $P = 0.00036$ on day 30). Previous reports have suggested that early after HCT, circulating MAIT cells are graft-derived (36). Thus, in recipients of BM grafts [which contain 12 times fewer mature T cells than PBSC grafts (39)], it is to be expected that there are fewer MAIT cells early after transplantation. In the PBSC graft recipients, above-median diversity was associated with higher MAIT cell frequency at day 30, an effect that was not observed in the BM graft recipients (Fig. 1C, $P = 0.014$). More than half (63%) of the BM graft recipients received GVHD-prophylaxis regimens containing posttransplantation cyclophosphamide (PTCy), which was previously described to have a negative influence on MAIT cell reconstitution (36). Fecal peri-engraftment α -diversity was measured as the median Simpson reciprocal index among fecal samples collected between days 7 and 21 for each patient, a time period we have previously shown to be predictive of clinical outcomes in allo-HCT patients (11). In addition to performing analyses in a dichotomous fashion dividing by median diversity of the cohort, we analyzed fecal α -diversity and MAIT cell frequency as continuous variables in the PBSC recipients where we had observed a difference. We confirmed a correlation between early MAIT cells and fecal diversity on day 30 (Fig. 1D, $R = 0.25$, $P = 0.0063$). This relationship between day 7 and day 21 fecal diversity and MAIT cells was not observed when MAIT cell frequencies were assessed at day 100, but the patient sample size is smaller at this latter time point (Fig. 1E).

MAIT cell frequencies did not appear to be dependent on pretransplantation conditioning intensity, with equivalent results seen in patients receiving myeloablative, reduced intensity, and nonablative pretransplantation conditioning regimens (fig. S7). Multivariate analysis was performed to evaluate the association of other potential confounding variables with MAIT cell frequency on day 30. Although the PTCy-based GVHD prophylaxis regimens had a negative association with MAIT frequency, peri-engraftment microbial diversity was confirmed as a significant independent factor associated with early MAIT cell frequency (Table 2).

Next, we sought to examine the relationship between MAIT cell frequency and patient outcome. We observed that patients who received PBSC grafts ($n = 118$) with higher-than-median MAIT cell frequency on day 30 (median = 0.64% of CD3⁺ cells) had a significantly increased overall survival (Fig. 1F, $P = 0.047$) and decreased nonrelapse mortality (Fig. 1G, $P = 0.031$).

To assess whether MAIT cell frequency was a predictor of GVHD, we excluded any patient who developed GVHD before blood sampling ($n = 115$ remaining in the cohort), and we observed a clear association between MAIT cell frequency and the subsequent development of grades 2 to 4 aGVHD and lower gastrointestinal (GI) GVHD in patients receiving PBSC grafts ($n = 89$) but not in patients receiving BM grafts ($n = 26$); (Fig. 1, H and I, $P = 0.0047$ and $P = 0.031$, respectively).

Frequency of Vδ2⁺ T cell subsets correlates with MAIT cells in high-dimensional flow cytometry analysis of early posttransplantation blood samples

Our interest extended beyond the MAIT cells, and our complete flow cytometry panel consisting of 22 markers defining conventional and unconventional T cell populations (table S1) allowed characterization of a number of different cell types in these post-HCT samples. The analysis was performed on samples collected from patients who received PBSC grafts, excluding patients who developed GVHD

Table 2. Multivariate analysis of factors associated with MAIT cell frequency (day 30).

Variable		Estimate (95% CI)	P value
Peri-engraftment α -diversity*	Continuously	0.178 (0.039–0.317)	0.013
PTCy	No versus yes	0.62 (0.309–0.932)	<0.001
Conditioning intensity	Nonablative versus ablative	–0.084 (–0.414–0.247)	0.62
	Reduced intensity versus ablative	–0.192 (–0.429–0.045)	0.114
Age	Continuously	–0.003 (–0.012–0.006)	0.58
Donor age	Per 10 years	–0.058 (–0.118–0.002)	0.059
HCT-CIT	2–3 versus 0–1	–0.105 (–0.346–0.135)	0.393
	>3 versus 0–1	–0.252 (–0.501–0)	0.051

*Median was calculated if a patient had more than one sample. †Hematopoietic cell transplantation comorbidity index (77).

before their blood sample was collected and those who received PTCy or had extremely low CD3 counts ($n = 76$ remaining in the cohort). We divided patients into higher-than-median MAIT cells or lower-than-/equal-to-median MAIT cells (median = 0.84% of CD3⁺ cells). We first performed a Uniform Manifold Approximation and Projection (UMAP) analysis to examine the general distribution of subpopulations within the CD3⁺ T cell compartment and color-coded the cells blue or red, according to whether the sample contained above- or below-median MAIT cell frequencies, respectively ($n = 38$ in each group; Fig. 2A). To annotate the CD3⁺ T cell clusters seen in our UMAP plot in these two groups of patients, we first used a traditional gating strategy (Fig. 2B). Using this approach, we could identify two populations that distinguished the two groups in Fig. 2A. The first was the MAIT cells themselves (Fig. 2A, rectangle), as expected. Unexpectedly, another non-MAIT cell cluster was also enriched in samples with highly abundant MAIT cells (Fig. 2A, circle). This cluster was identified as the Vδ2 subset of the $\gamma\delta$ T cells (Fig. 2C, circle). When quantified as a frequency of the CD3⁺ cells in this patient cohort, we observed a higher Vδ2 cell frequency in the patients with above-median MAIT cell frequency (Fig. 2D, $P = 0.004$). To further explore and validate the differences between patients with higher and lower MAIT cell frequency, we defined phenotypic clusters using the self-organizing maps (FlowSOM) (40) algorithm (Fig. 2E), in which 19 markers were used to define 20 different clusters within the CD3⁺ compartment. When examining each of the 20 populations defined by the FlowSOM algorithm, we observed four clusters that were more frequent in the above-median MAIT cell group. Cluster 5 corresponded to a cell population expressing MAIT cell markers (MR1⁺, Vα7.2⁺, and CD161⁺), as expected on the basis of how the patient samples were grouped. Two other clusters (clusters 8 and 14) identified subpopulations highly expressing the Vδ2 cell marker (Fig. 2, F and G). The fourth cluster

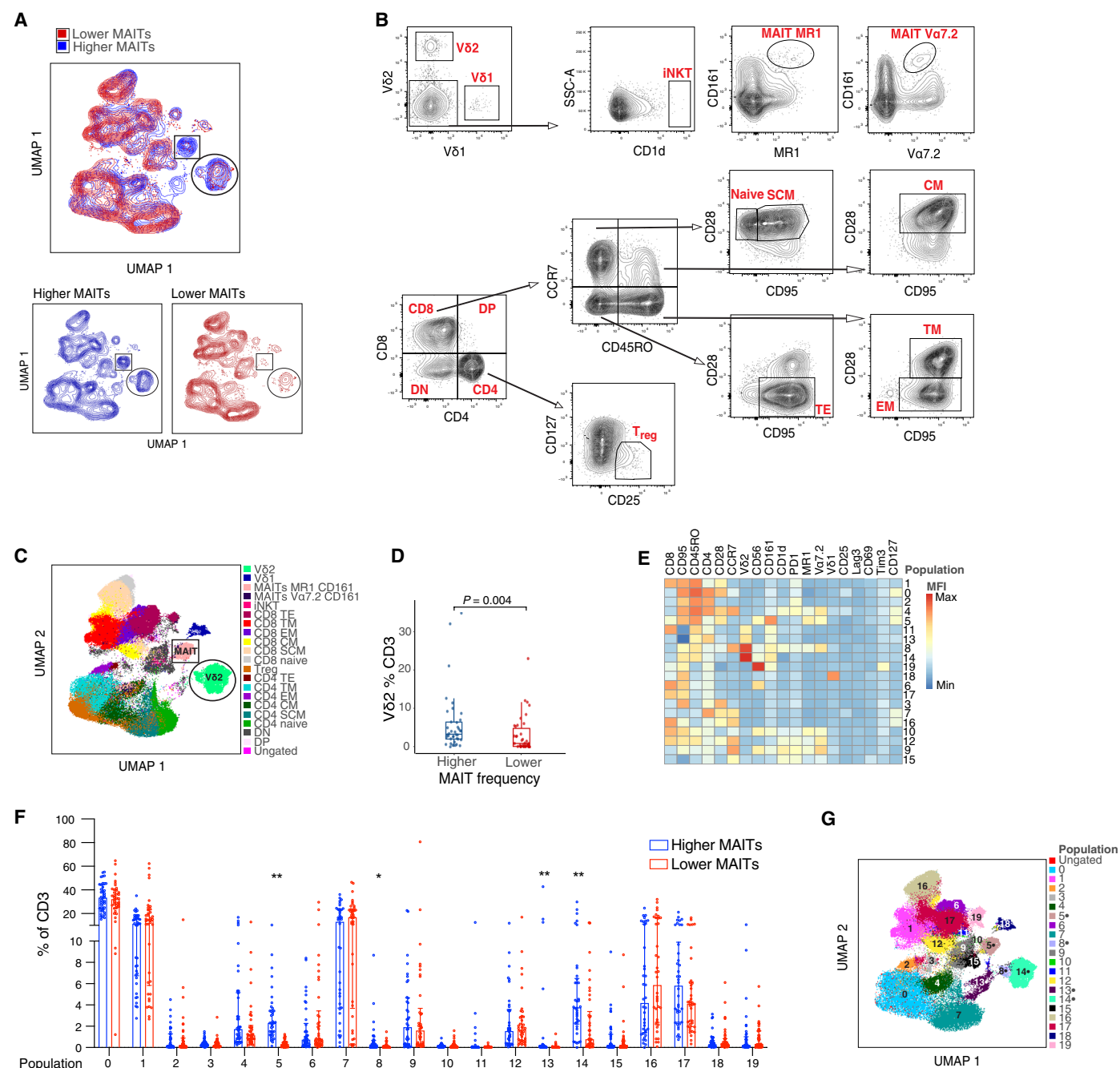


Fig. 2. Deep immune profiling reveals higher frequency of the Vδ2 subset of γδ T cells in the samples with higher MAIT frequency after HCT. (A) UMAP clusters in patients with day 30 MAIT frequency higher than (blue) or lower than/equal to (red) the population median. Populations different between the two groups are marked by a circle and a rectangle. (B) Gating strategy for both the conventional and unconventional T cell populations, which were used to define the UMAP clusters in (A). Concatenated CD3⁺ cells from the same 76 samples from (A) were used to establish the gating scheme. DN, double negative; DP, double positive; SCM, stem cell memory; CM, central memory; TM, transitional memory; EM, effector memory; TE, terminal effector. (C) UMAP clustering color-coded to identify the gated populations. Clusters that were different between samples with higher and lower MAIT frequency are outlined: MAIT cells (rectangle) and Vδ2 cells (circle). (D) Vδ2 cell frequency as a function of higher or lower MAIT frequency (Wilcoxon rank-sum test, $n=76$, $P=0.004$). Each dot represents a single patient, boxes represent median with interquartile range. (E) Heatmap of 20 populations identified by self-organizing maps (FlowSOM) algorithm, demonstrating MFI of 19 identifying surface markers. (F) Frequency of each FlowSOM population in each patient sample represented by 3000 concatenated CD3⁺ events per sample. Each dot represents a single patient, and lines represent median with interquartile range. Mann-Whitney test was used to calculate differences between the two groups for each population, and FDR correction was performed for multiple hypothesis testing, total $n=76$, * $P<0.05$, ** $P<0.01$. (G) UMAP projection of 20 populations of CD3⁺ cells identified by the FlowSOM clustering algorithm. Dots indicate clusters that were different between samples with higher and lower MAIT frequency.

(cluster 13) had a mixed phenotype of CD4 expression and intermediate CD45RO, Vδ2, and CD161 expression. UMAP and FlowSOM analysis of 15 healthy volunteer samples demonstrated different clusters in this sample cohort than those seen in the patient samples (fig. S8, A to C), which can be explained given general changes in the T cell pool after allo-HCT and the timeline by which allo-HCT patients reach normal T cell counts and phenotypes.

Vδ2 cell frequencies correlate with intestinal microbial diversity and predict aGVHD

Having found that among 20 phenotypically unique CD3⁺ cell populations, only the Vδ2⁺ clusters correlated with MAIT cell frequency, we next examined the Vδ2 cell subset in all patients receiving PBSC allografts ($n = 118$). We confirmed the result from the patient cohort used for high-dimensional flow cytometry that the frequency of Vδ2 cells is correlated with the frequency of MAIT cells (Fig. 3A, $R = 0.38$, $P = 2.8 \times 10^{-5}$). Absolute counts of Vδ2 cells were lower at day 30 after HCT than in healthy volunteers; however, their frequency within the CD3 compartment was comparable (Fig. 3B). As we observed for MAIT cells, the day 30 Vδ2 cell frequency is significantly lower in recipients of BM grafts than PBSC grafts. However, unlike in the MAIT population, this difference persists until day 100 (fig. S9, A and B). We had demonstrated that patients with higher

MAIT proportion have higher peri-engraftment stool α -diversity (Fig. 1, C and D), and Vδ2 cells are known to depend on the microbial metabolite HMBPP (4, 5). We observed a significantly higher Vδ2 cell proportion in patients with higher-than-median peri-engraftment stool α -diversity compared with those with lower-than-/equal-to-median diversity (Fig. 3C, $P = 0.0026$). We confirmed a positive association between early Vδ2 cells and fecal diversity when fecal α -diversity and Vδ2 proportion were analyzed as continuous variables (Fig. 3D, $R = 0.22$, $P = 0.017$). When focusing on patients who developed GVHD on the day of or after blood sampling, we observed a negative association between Vδ2 cell frequency and the development of grades 2 to 4 aGVHD (Fig. 3E, $P = 0.013$) and lower GI GVHD (Fig. 3F, $P = 0.0099$; absolute counts shown in fig. S10, A to C), suggesting that these cells may play a protective role with respect to aGVHD development.

We performed flow cytometric analysis on additional day 30 samples from allo-HCT patients treated at another center [Duke University Medical Center; $n = 69$ peripheral blood mononuclear cell (PBMC) samples, $n = 58$ with matched stool samples, $n = 49$ received PBSC grafts], and we confirmed an association between the frequency of MAIT and Vδ2 populations. However, in this small cohort, we could not establish any additional associations (fig. S11, A to E).

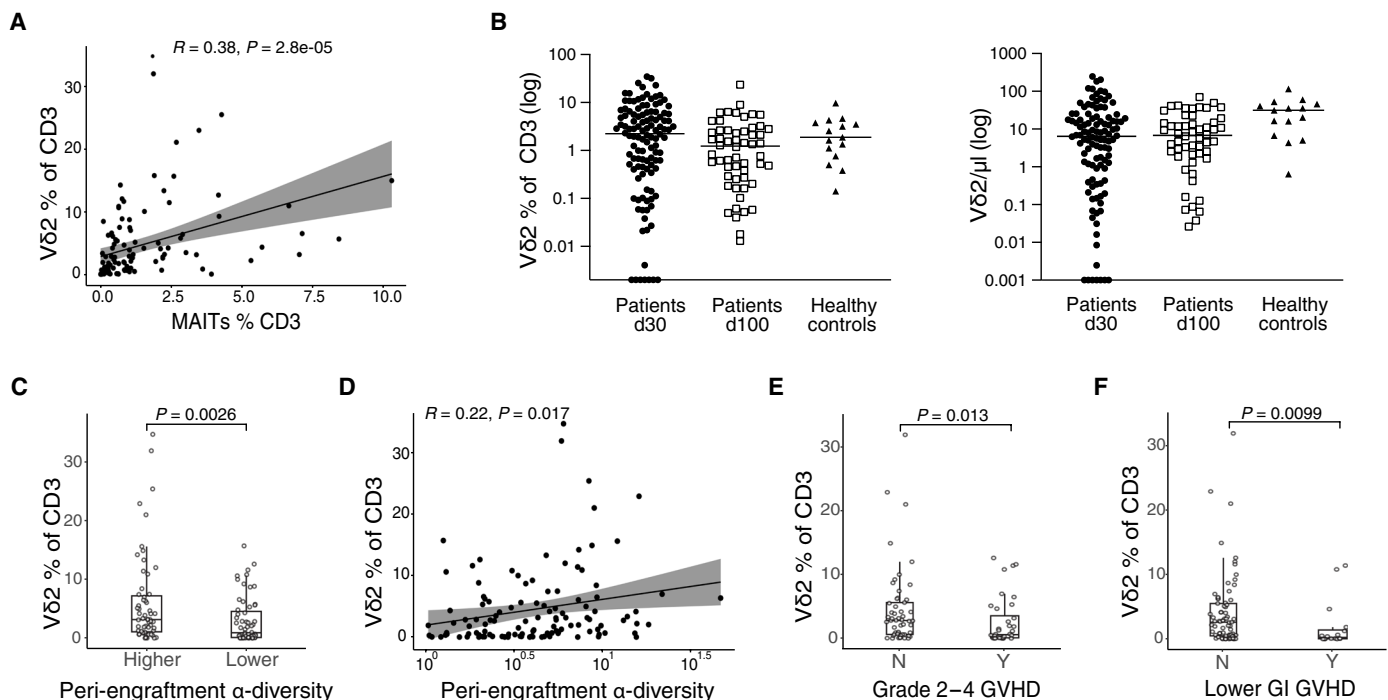


Fig. 3. The frequency of the Vδ2 subset of $\gamma\delta$ T cell is associated with higher fecal α -diversity in peri-engraftment stool samples and lower rates of overall and lower intestinal acute GVHD. (A) MAIT and Vδ2 cell frequency analyzed as continuous variables in day 30 blood samples from recipients of PBSC grafts (Pearson correlation, $n = 118$, $R = 0.38$, $P = 2.8 \times 10^{-5}$). (B) Vδ2 cell frequencies among CD3⁺ cells and absolute counts on day 30 (± 10) and day 100 (± 20) after HCT in PBSC graft recipients (frequencies: day 30 $n = 118$, day 100 $n = 56$; healthy $n = 15$; absolute counts: day 30 $n = 110$, day 100 $n = 54$, healthy $n = 15$). Each symbol represents a single patient sample, lines represent median. (C) PBSC graft recipients were classified as in Fig. 1C into $>$ median (higher) and $\leq$ median (lower) α -diversity, and Vδ2 cell frequency in each subgroup at day 30 was compared (Wilcoxon rank-sum test, $n = 118$, $P = 0.0026$). (D) Fecal α -diversity between d7 and d21 as a continuous variable correlated with Vδ2 frequency (Pearson correlation, $n = 118$, $R = 0.22$, $P = 0.017$). (E) PBSC graft recipients were classified by GVHD grade (grades 2 to 4, Y = Yes, grades 2 to 4 GVHD present, N = No, grades 2 to 4 GVHD) and Vδ2 frequencies compared in each group (Wilcoxon rank-sum test, $n = 53$ grades 0 and 1 GVHD, $n = 36$ grades 2 to 4, $P = 0.013$). (F) Patients were classified by lower GI GVHD status (Y = Yes, lower GI GVHD present or N = No, no lower GI GVHD) and Vδ2 frequencies were compared in each group (Wilcoxon rank-sum test, $n = 73$ without lower GI GVHD, $n = 16$ with any stage lower GI GVHD, $P = 0.0099$). (E and F) Patients with blood samples collected before or on the day of GVHD onset were included. (A and D) Each dot represents a single patient sample. (C, E, and F) Each dot represents a single patient, and boxes represent median with interquartile range.

Specific bacterial taxa and their metabolites support MAIT and Vδ2 cell reconstitution

To extend our microbiota-related findings beyond α -diversity and further examine the microbial community profiles associated with MAIT and Vδ2 cell reconstitution, we used linear discriminant analysis of effect size (LEfSe) analysis (41) to identify a number of microbial taxa that are associated with higher-than-median or lower-than-/equal-to-median MAIT cell frequency at day 30 after transplantation. A higher abundance of the phylum *Bacteroidetes* was associated with higher MAIT cell frequency at day 30, whereas the phylum *Firmicutes* is more abundant in the patient with lower MAIT cell frequency (Fig. 4, A and B), corresponding with an in vitro study of the ability of human commensal strains to stimulate an engineered MAIT reporter system (42). Moreover, other members of this phylum, *Bacilli* (class) and *Lactobacillales* (order), the abundance of which was inversely correlated with MAIT cell frequencies, are predominantly composed of species such as *Enterococcus*, which lack the riboflavin biosynthesis pathway and therefore do not synthesize the MAIT-activating metabolites (43). Conversely, higher MAIT cell frequency was associated with a higher relative abundance of the class *Bacteroidia* and order *Bacteroidales*, as well as the class *Erysipelotrichia*, order *Erysipelotrichales*, family *Erysipelotrichaceae*, and the genus *Erysipelatoclostridium*. The bacterial family *Ruminococcaceae* and genus *Blautia* were also more abundant in patients with higher MAIT frequency, the latter in concordance with previously published findings in the allo-HCT patients (36). The differences observed in the LEfSe analysis were further confirmed in the comparison of the relative abundance of each taxon on a per-patient basis (Fig. 4, C and D).

As the specific bacterial ligands required for MAIT and Vδ2 cell activation and proliferation are known, we hypothesized that we would be able to identify the pathways associated with the production of these ligands by using the Phylogenetic Investigation of Communities by Reconstruction of Unobserved States (PICRUSt2) (44) algorithm. This package uses 16S amplicon abundance data with data from published metagenomes to predict the abundance of functional pathways. We investigated whether the higher frequency of MAIT cells in the day 30 blood samples is associated with the higher abundance of the riboflavin biosynthesis pathway, which leads to the production of activating MAIT cell ligands. Using the Kyoto Encyclopedia of Genes and Genomes (KEGG) orthology analysis (45) and the MetaCyc pathway database (46), we showed that the abundance of both the riboflavin biosynthesis pathway (Fig. 4E) and the key enzymes belonging to this pathway, as previously reported (47) (Fig. 4, F and G), was not differentially present in the patients with higher versus lower MAIT cell frequency. These findings are consistent with an uncoupling of production capacity and true production, as has been demonstrated in vitro (42).

In contrast to these findings, the microbial gene abundance of the MEP biosynthesis pathway, resulting in the production of Vδ2 cell activating ligand HMBPP was significantly more abundant in the samples from patients with Vδ2 cell frequency higher than the median (median Vδ2 frequency = 2.235% of CD3⁺ cells) when compared with those with the frequency equal to or lower than the median (Fig. 4I, $P = 0.00054$). Several bacterial species that synthesize isoprenoids using the mevalonate pathway and not the MEP pathway [e.g., *Enterococcus* and *Streptococcus* (48)] belong to the order *Lactobacillales*, which was markedly increased in the patients with lower MAIT and Vδ2 cell frequency. The higher predicted MEP

pathway abundance in Vδ2-high patients was further confirmed by higher gene expression of several key enzymes of this pathway in the patient stool samples (Fig. 4, J and K).

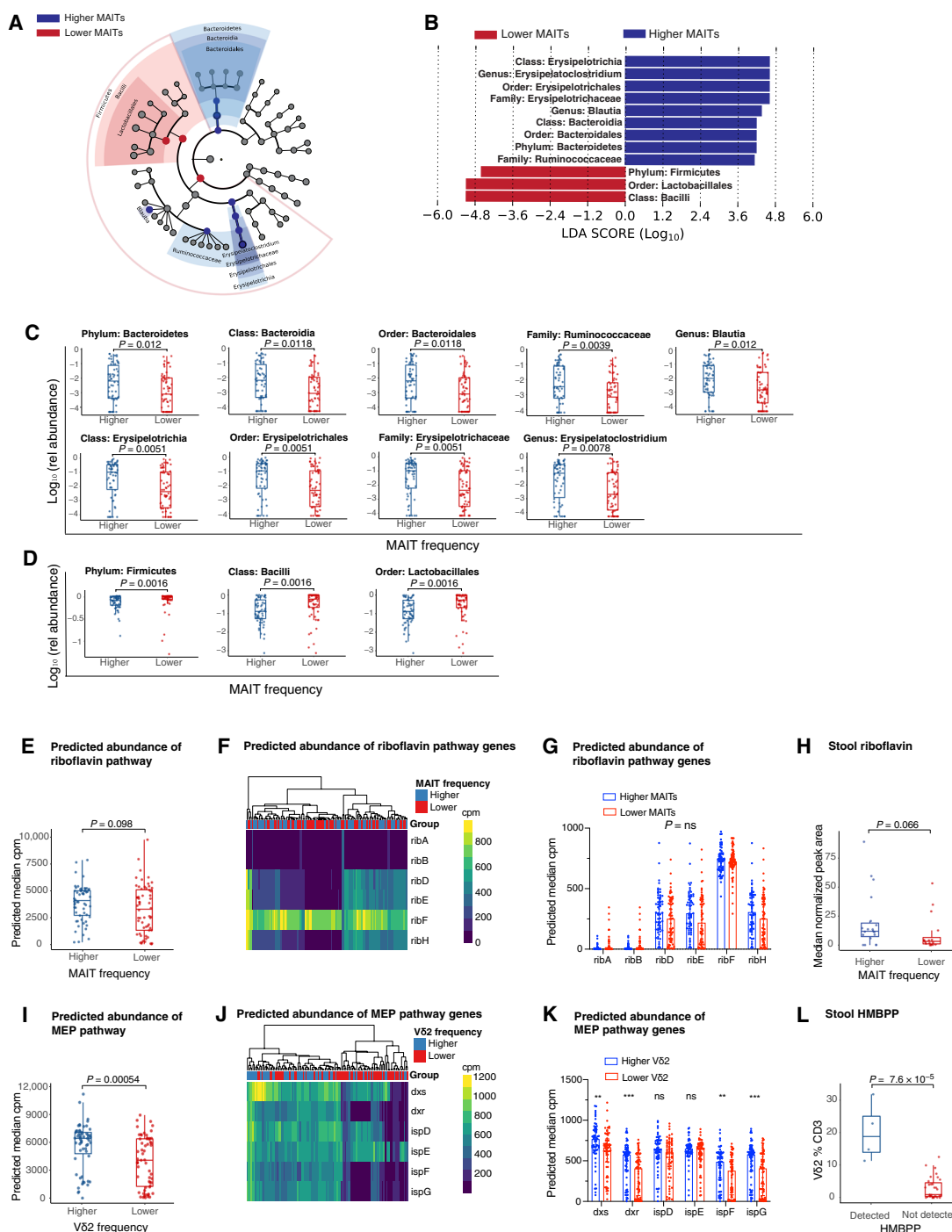
We next performed metabolomic analysis of stool samples from patients with high and low MAIT and Vδ2 cell frequencies (19 patients with the highest and 16 with the lowest MAIT cell frequencies; $n = 70$ peri-engraftment samples available; 1 to 3 samples per patient). Attempts to measure previously described MAIT ligands in human stool extracts were unsuccessful, likely because of the low concentration and reactive nature of these metabolites. However, we observed a trend toward a higher riboflavin peak area in patients with above-median MAIT cell frequency (Fig. 4H, $P = 0.066$). Multiple activating ligands for MAIT cells are derived from 5-amino-6-(D-ribitylamino)uracil, an intermediate in the riboflavin biosynthetic pathway. Therefore, although riboflavin per se is not an activating ligand for MAIT cells, its concentration may reflect the capacity of intestinal microbial communities to produce the other less stable MAIT cell ligands. In contrast, MEP pathway intermediate HMBPP is one of the most potent activating ligands for Vδ2 cells with an EC₅₀ (median effective concentration) < 1 nM (49). This metabolite was detected in stool extracts from only four patients in our cohort. These patients had very high Vδ2 cell frequencies in relation to the rest of our patient population (Fig. 4L). HMBPP was below the limit of detection in the stool extracts from the remaining patients analyzed. Together, these results provide evidence that microbial metabolic products responsible for the activation of both MAIT and Vδ2 cells are present in stool samples from patients with allo-HCT and, when detectable, correlate with an increased frequency of these unconventional T cells.

MAIT and Vδ2⁺ T cell subsets acquire effector/cytotoxic transcriptional signature post-HCT

Having observed an association between higher MAIT and Vδ2 T cells and favorable transplant outcome, we examined the transcriptional landscape of these cells using single-cell RNA sequencing (scRNA-seq) analysis of fluorescence-activated cell sorter (FACS)-purified MAIT and Vδ2 cell populations. We selected five patients with a good transplant outcome that was not complicated by GVHD or early relapse, who had high frequencies of both MAIT and Vδ2 T cells and available PBMC samples. We compared FACS-purified MAIT and Vδ2 populations from allo-HCT patients with cells from healthy volunteers. We identified a number of transcriptional features that are more pronounced in the post-HCT period. MAIT cells up-regulated genes linked with effector functions and cytotoxicity, such as *GNLY*, *PRF1*, and *CCL4*, as well as migration/invasiveness, such as the adhesion molecule *ITGB2* and chemokine receptor *CD74*. Genes encoding negative feedback inhibitors of the nuclear factor κ B (NF κ B) pathway (*NFKBIA* and *TNFAIP3*) were down-regulated, further suggesting the proinflammatory capacity of post-HCT MAIT cells (Fig. 5A). Post-HCT Vδ2 cells appeared to share many transcriptional changes with MAIT cells, e.g., up-regulation of *GNLY*, *PRF1*, and *CCL4*, as well as granzymes *GZMB* and *GZMH* (Fig. 5B).

It has been suggested that MAIT cells can contribute to tissue repair (7, 50, 51). Thus, we used a reported gene set defining a tissue repair signature and performed a Gene Set Enrichment Analysis (GSEA) (52–54). Most of these genes were absent in our dataset or were expressed at a low level, and the pathway was not enriched in patients compared with healthy controls (fig. S12, A to D).

(A) Differences in peri-engraftment stool microbiota composition between PBSC recipients with MAIT cell frequency $>$ median (higher) or $\leq$ median (lower) of all patients depicted using LEfSE ($n = 425$ samples collected on d7 to d21 after HCT from 118 patients; median relative abundance per patient at genus level). (B) Dominant taxa driving the difference between the two groups (LDA > 4 , $P < 0.01$). (C) Taxa dominant in the higher MAIT cell group from the LEfSE analysis plotted as a relative abundance in patients with higher versus lower MAIT cells as defined above. (D) Taxa dominant in the lower MAIT cell group identified by the LEfSE analysis plotted as a relative abundance in patients stratified by MAIT cell frequency. (C and D) Wilcoxon rank-sum test was performed to compare taxa abundances between groups, FDR correction was performed for multiple hypothesis testing. (E) Predicted gene abundance of the riboflavin biosynthesis pathway in PBSC recipients as a function of MAIT cell frequency (Wilcoxon rank-sum test, $n = 118$, $P = 0.098$). (F) Heatmap representing microbial gene level abundance of key riboflavin biosynthesis enzymes in patients with higher and lower MAIT cell frequency. (G) Differences in predicted microbial gene abundance of specific genes encoding key riboflavin biosynthesis enzymes. (H) Metabolomic analysis was performed on stool samples collected from patients with previously defined higher or lower MAIT cells. Riboflavin concentrations measured as peak area normalized to internal standard, median value was used when more than one sample per patient was analyzed (Wilcoxon rank-sum test, $n = 35$, $P = 0.066$). (I) Predicted microbial gene analysis (Wilcoxon rank-sum test, higher and lower V δ 2 cell frequency cohort as in (H). Patients were divided into two groups ($P < 10^{-5}$). (E to G and I to K). A total of 1000 genes were analyzed. (J) Relative abundance in all samples available for each gene. (G to I, and K and L) Each dot represents a patient, color represents the enzyme, and FDR correction was performed for multiple hypothesis testing.



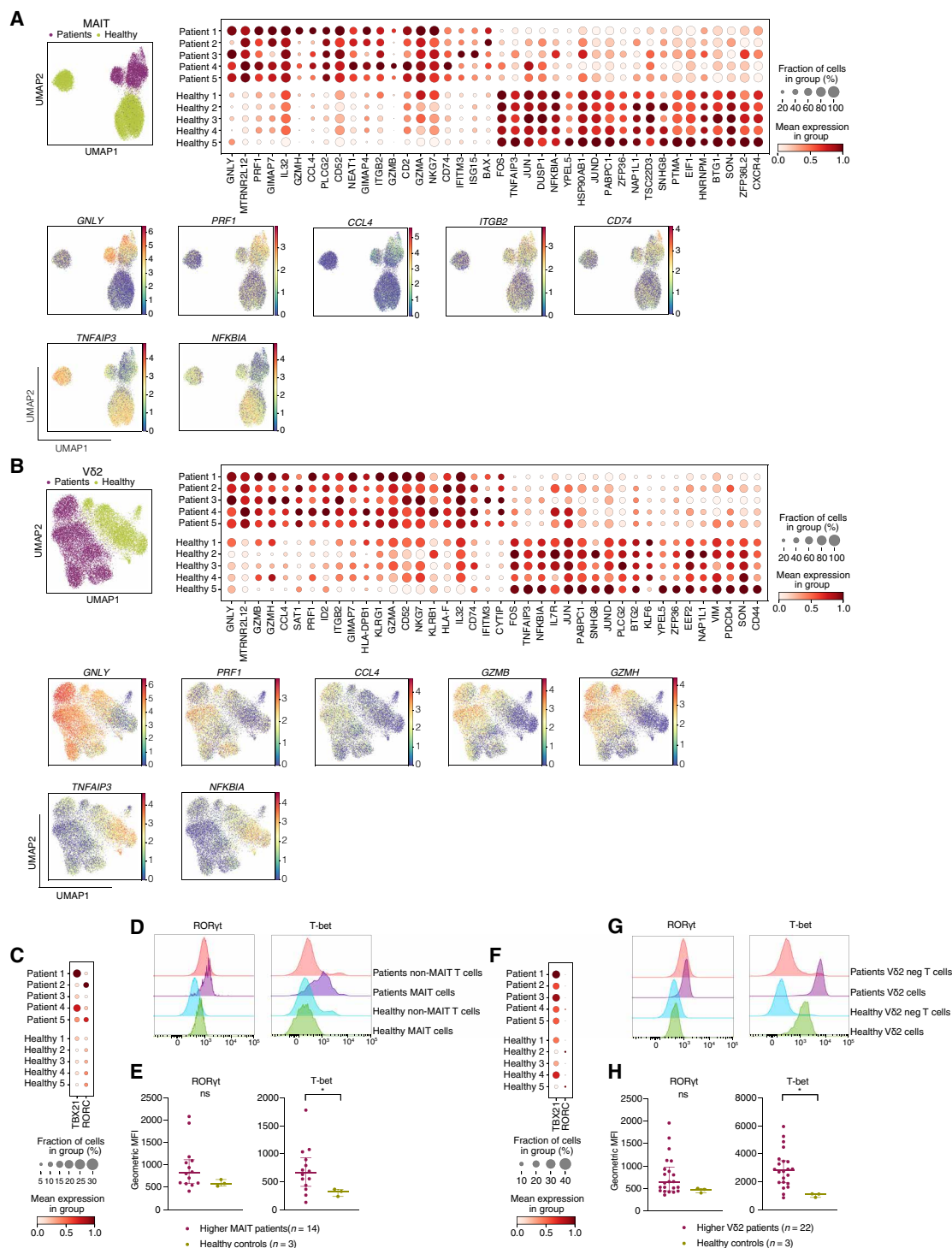


Fig. 5. MAIT and Vδ2⁺ T cell subsets acquire effector cytotoxic transcriptional signature posttransplantation and are mostly Th/Tc1 polarized.

(A) Top left: UMAP clustering of MAIT cells from patients ($n=5$) and healthy donors ($n=5$). Top right: Dot plot demonstrating the most differentially regulated genes between patients and healthy donors. Bottom: UMAP plots of the most differentially regulated genes between healthy and patient MAIT cells. (B) Top left: UMAP clustering of Vδ2 cells from patients and healthy donors. Top right: Dot plot quantifying the most differentially regulated genes between patients and healthy donors. Bottom: UMAP plots of most differentially regulated genes between healthy and patient Vδ2 cells. (C) Gene expression of T-bet and RORγt in patients ($n=5$) and healthy controls ($n=5$). (D and E) Representative histograms and quantification of RORγt and T-bet expression in MAIT cells from patients with previously defined above-median MAIT cell frequency ($n=14$) and healthy controls ($n=3$) by flow cytometry. Geometric MFI=geometric mean fluorescence intensity. (F) Gene expression of T-bet and RORγt in Vδ2 cells in patients ($n=5$) and healthy controls ($n=5$). (G and H) Representative histograms of RORγt and T-bet expression in Vδ2 populations from PBMC samples of patients with previously defined above-median Vδ2 cell frequency ($n=22$) and healthy controls ($n=3$) by flow cytometry. (E and H) Each dot represents a single patient; lines represent median with interquartile range. Mann-Whitney test was used to compare the groups, $*P < 0.05$.

Human peripheral blood MAIT cells can express typical Th/Tc1 transcription factor T-bet and Th/Tc17-related transcription factor RORγt. When activated, circulating MAIT cells are thought to acquire a Th/Tc1 phenotype, with most secreting tumor necrosis factor (TNF) and interferon γ (IFN γ) and only a small subset producing interleukin-17 (IL-17) (55). In contrast, MAIT cells in granulocyte colony-stimulating factor–mobilized PBSC grafts were shown to bear a Th/Tc17 phenotype, characterized by

RORγt up-regulation (56). We observed an increase in T-bet expression at both the RNA and protein levels in patient samples when compared with the healthy volunteers (Fig. 5, C to E). Post-HCT Vδ2 cells also expressed more T-bet than this population at steady state (Fig. 5, F to H). Furthermore, a higher proportion of MAIT expressed Ki-67 after transplantation, suggesting increased proliferation after allo-HCT compared with the steady state (fig. S13).

DISCUSSION

Our study of patients with allo-HCT supports the notion that MAIT and V δ 2 cells depend on microbe-derived factors and that, together, they support favorable posttransplantation outcomes. Prior work has explored the relationship of MAIT cells with a favorable allo-HCT outcome (33, 36–38) and suggested that early V δ 2 abundance is associated with lower rates of GVHD (30), but here, we link an in-depth analysis of posttransplant immunophenotype with scRNA-seq, metabolomics, and large-scale microbiome profiling. The association between commensal microbiota and MAIT cells has been explored in the clinical HCT setting and also in different states and diseases in both mouse and human studies (7, 20, 36–38, 57). The relationship of $\gamma\delta$ T cells with the commensal microbiota has not been addressed in great detail, and the relationship between the gut microbiome and the V δ 2 cell subset of $\gamma\delta$ T cells has not been described to date. We have demonstrated that both MAIT and V δ 2 cells are present in markedly higher numbers in patients with higher fecal α -diversity. Both MAIT and V δ 2 cells require specific bacterial metabolites for development and maintenance. In our metabolomic analysis, we observed a trend toward higher riboflavin concentrations in stool samples from patients with higher MAIT cells, although this did not reach statistical significance with the number of samples analyzed. The V δ 2 cell-activating metabolite HMBPP could be measured in a small number of our samples, and consistent with the PICRUSt2 data, these were from patients with the highest frequencies of circulating V δ 2 cells. Although other ligands of V δ 2 cells have been described, such as isoprenyl pyrophosphate, HMBPP is by far the most potent activator of human V δ 2 T cells (58). Given its activity at nanomolar concentrations (49), it is possible that this metabolite is capable of influencing V δ 2 cell frequencies at concentrations below the detection threshold of our method.

The mechanism by which donor MAIT and V δ 2 cells may protect against aGVHD is not yet clear. MAIT cells are best known for their production of cytotoxic mediators and cytokines upon activation, such as perforin and granzymes, IFN γ , TNF, and IL-17, although the latter is not thought to be abundantly produced by circulating MAIT cells at steady state (55). Moreover, MAIT cells in the skin have been associated with tissue repair in a mouse model (7). In the context of GVHD, a preclinical study demonstrated that recipient MAIT cells are protective with regard to acute gut GVHD in an IL-17-dependent fashion (20). Although less is known regarding the V δ 2 T cells because this specific $\gamma\delta$ population cannot be studied in mice, V δ 2 cells can produce IFN γ , and a subset of the V δ 2 cells of healthy neonates and adults was also shown to produce IL-17 when stimulated *in vitro* with IL-6, IL-23, IL-1 β , and transforming growth factor- α (59, 60). Our data suggest that the MAIT and V δ 2 populations undergo comparable transcriptional changes after allo-HCT, consistent with a gain of cytotoxic and effector function. Although we observed expression of some genes associated with tissue repair in MAIT and V δ 2 cells, their expression was low, and this pathway was not up-regulated in the patients compared with healthy volunteers. The up-regulation of the proinflammatory and cytotoxic genes in microbiota-responsive populations could reflect a role for these populations in controlling pathogenic bacteria or cell populations. Recent studies have indeed shown that granulysin facilitates granzyme delivery into bacterial cells and causes their death (61, 62). Furthermore, both MAIT and V δ 2 cells have been demonstrated to release granulysin and perforin upon TCR activation in the context of infection (63, 64). These hypotheses will require further

exploration in prospective studies, and moreover, circulating unconventional T cells might differ from their tissue-resident counterparts.

We are cognizant that there are several limitations of our study. We assessed the association between the intestinal microbiome and MAIT and V δ 2 cells in samples that were collected and available as part of broad biobanking efforts at our center rather than conducting a prospective, multicenter study. We believe that multicenter studies and deeper mechanistic analyses will be needed to validate the positive predictive effects of these populations on the outcome of allo-HCT patients. Although our metabolomic analysis corroborates the PICRUSt2 predictions, we could not measure metabolites in all stool samples from our patient cohort, and thus, the robustness of this analysis remains limited by small patient numbers.

In summary, we propose that microbiome-derived molecules support the maintenance of both MAIT and V δ 2 cell populations after allo-HCT. Strategies to support intestinal microbial diversity may therefore be beneficial for transplantation patients because of a positive effect on unconventional T cell populations.

MATERIALS AND METHODS

Study design

The objective of the study was to examine the association of post-transplantation unconventional T cell subsets, their relationship with conventional T cells, and intestinal microbiome composition and characteristics in patients undergoing allo-HCT. Inclusion criteria for this study were as follows: patients undergoing allo-HCT with an unmodified PBSC or BM graft at Memorial Sloan Kettering Cancer center (MSKCC) in New York with an evaluable PBMC sample collected on day +30 (± 10 days) and/or day +100 (± 20) relative to HCT and at least one available stool sample (successfully 16S-amplified and sequenced with >200 reads) that had been collected between days +7 and +21. One patient who received a combination of BM and PBSC grafts and one patient who did not engraft were excluded from the study. Inclusion and exclusion criteria are described in full in fig. S1. Five of the patients in the cohort had participated in a randomized clinical trial of fecal microbiota transplantation (FMT, NCT02269150) (65) and received FMT after stool and blood samples were collected for these analyses. Clinical aGVHD was graded according to the Glucksberg criteria and reviewed in a consensus meeting at our institution (66).

The biospecimen collection protocol was approved by the Institutional Review Board (IRB) at MSKCC (protocol numbers: 06-107 and 16-834), and written informed consent was obtained from all participants. Control blood samples were collected from healthy volunteers who also provided a written informed consent to the biospecimen collection protocol approved by the IRB at MSKCC.

Stool collection and analysis, taxa identification using LEfSe, and metagenomic prediction

Stool samples were collected in the peri-engraftment time window, defined as days +7 to +21 after HCT. DNA extraction, polymerase chain reaction amplification of genomic 16S ribosomal RNA V4-V5 regions, and sequencing were performed as previously described (11–13).

The α -diversity was calculated using the reciprocal Simpson index on the amplicon sequence variant (ASV) level. The 16S sequencing data were examined with the DADA2 pipeline (67) using default parameters except for the maxEE = 2 and truncQ = 2 in filterandtrim() function. For every sample, read number was capped

at 100,000. ASVs were annotated on the basis of the National Center for Biotechnology Information 16S ribosomal RNA sequences database using BLAST (68). The α -diversity was calculated using the reciprocal Simpson index on the ASV level.

LEfSe is a high-dimensional class comparison algorithm that performs nonparametric statistical testing as well as additional tests to assess biological relevance, to describe taxonomic features leading to major differences between two groups (41). Genus-level abundances were analyzed by LEfSe using Galaxy server (<https://huttenhower.sph.harvard.edu/galaxy/root>). Only the taxa with relative abundance higher than 10^{-4} in at least 10% of the samples were included. For patients with multiple fecal samples in the sampling window, median genus-level abundances were aggregated for analysis. Differentially abundant taxa between the two patient groups with higher and lower MAIT cell frequency were considered statistically significant using the P value <0.01 and local density approximation (LDA) log score cutoff ± 4 .

PICRUSt2 is a method used to predict functional characteristics of bacterial communities based on 16S sequencing (44). ASV counts for each sample and the sequences for each unique ASV were used to run through the PICRUSt2 pipeline. Gene families were annotated against the KEGG orthology (45). The MetaCyc database was used to identify the specific pathways (46). The predicted abundance for a KEGG ortholog or a pathway per patient were calculated as a median abundance of this gene or pathway across all samples collected from a given patient in the peri-engraftment period expressed in copies per million. The pathway and enzyme identifiers from the KEGG orthology and the MetaCyc database used in the analysis are listed in table S2.

Metabolomic analyses

Metabolite extraction

Stool samples (180 to 200 mg) were weighed into bead tubes (Omni) for homogenization. Samples were then extracted using nine volumes of liquid chromatography–mass spectrometry (LC-MS) grade 80% methanol containing 100 nM $^{13}\text{C}_4$ $^{15}\text{N}_2$ -riboflavin (IsoSciences) as an internal standard and homogenized using a BeadRuptor (OMNI) for 3 min at 6 m/s. After overnight incubation at -80°C , samples were vortexed and centrifuged at 20,000g for 20 min to clarify and precipitate protein. Extracts were then dried in a vacuum evaporator (Genevac EZ-2 Elite) for 2 hours and resuspended in 75 μl of 50% methanol. Last, samples were vortexed, incubated on ice for 20 min, and clarified by centrifugation at 20,000g for 20 min at 4°C . The supernatant was divided into two injection vials for acquisition with LC-MS tandem mass spectrometry (LC-MS/MS) methods (ion pair and reverse-phase LC separations).

LC-MS/MS metabolite detection

Ion pair LC-MS/MS analysis was performed with LC separation on a Zorbax RRHD Extend-C18 column (150 mm by 2.1 mm, 1.8- μm particle size, Agilent Technologies) and using a gradient of solvent A (10 mM tributylamine and 15 mM acetic acid in 97:3 water:methanol) and solvent B (10 mM tributylamine and 15 mM acetic acid in methanol) according to the manufacturer's instructions (MassHunter Metabolomics dMRM Database and Method, Agilent Technologies). LC separation was coupled to an Agilent 6470 series triple quadrupole system using a dual Agilent Jet Stream source. The capillary voltage was 2000 V, the nebulizer gas pressure was 45 psi, the drying gas temperature was 250°C , the drying gas flow rate was 13 liters/min, and the delta electron multiplier voltage (EMV) was 600 V. The injection

volume was 7.5 μl . HMBPP was detected in negative mode with the following transitions: 261 $\rightarrow$ 159 (CE 20); 261 $\rightarrow$ 79* (CE 28), *indicates primary transition used for quantitation.

Reverse-phase LC-MS/MS analysis for riboflavin detection was performed with LC separation on a CORTECS UPLC C18+ (150 mm by 2.1 mm, 1.6- μm particle size, Waters) and using a gradient of solvent A (8.7 mM formate/formic acid, pH 2.9 in water) and solvent B (100% acetonitrile). A linear chromatographic gradient was used (0.5% B to 99.5% B in 12 min) at 0.4 ml/min flow rate, followed by 2 min of flushing at 99.5% B at a flow rate of 0.5 ml/min. LC separation was coupled to an Agilent 6495C series triple quadrupole system using a dual Agilent Jet Stream source. The capillary voltage was 4000 V in positive and 3000 V in negative, the nebulizer gas pressure was 25 psi, the drying gas temperature was 230°C , the drying gas flow rate was 12 liters/min, and delta EMV was 600 V. The injection volume was 7.5 μl . Riboflavin was detected in positive mode with the following transitions: 377 $\rightarrow$ 57 (CE 32); 377 $\rightarrow$ 172 (CE 26); 377 $\rightarrow$ 243* (CE 24), *indicates primary transition used for quantitation. Data analysis was performed using MassHunter Quantitative Analysis (v. B.09.00).

PBMC isolation and flow cytometry

Patient PBMCs were collected into Becton Dickinson Vacutainer CPTM Cell Preparation Tubes with sodium citrate. Cells were frozen in liquid nitrogen in a freezing medium containing RPMI 1640, 10% dimethyl sulfoxide, and 12.5% human serum albumin. PBMCs from healthy samples were isolated using density centrifugation with Ficoll Paque. Between 0.3×10^6 and 10×10^6 PBMCs were stained in phosphate-buffered saline with LIVE/DEAD Zombie ultraviolet fixable dye (BioLegend) for 20 min at room temperature. Human Fc block (Miltenyi Biotec) was then added for 10 min at 4°C , followed by a staining with MR1-5-OP-RU APC and CD1d-PBS-57 BV421 or MR1-Ac-6-FP and CD1d-unloaded tetramer (43) (acquired from National Institutes of Health tetramer core facility) for 40 min at room temperature. The tetramer staining was followed by surface antibody staining for 15 min at room temperature. In each batch of samples (n per batch ranged from 23 to 30), we included one or more healthy control samples stained with loaded and unloaded tetramer. In most batches, we included a pooled patient sample stained with the unloaded tetramer.

For the intracellular flow cytometry analysis, the tetramer and surface staining was performed as described above. Cells were then fixed for and permeabilized with the Foxp3/transcription factor staining buffer set (eBioscience) according to the manufacturer's instructions. Intracellular antibody staining was performed for 30 min at room temperature.

The complete list of antibodies in the flow cytometry panel is available in table S1. Samples were analyzed using LSR Fortessa X50 Symphony cytometer (BD Biosciences). Standardized SPHERO rainbow beads (BD Biosciences) and SPHERO Supra Rainbow Beads (Spherotech) were used to track and adjust photomultiplier tube voltages over time. UltraComp eBeads (Invitrogen) were used for compensation.

Flow cytometry data were analyzed using FlowJo Software version 10.7.2 (FlowJo LLC, Ashland, OR). Patients who had no measurable lymphocyte or CD3 $^+$ population either in the complete blood count or in the flow cytometry analysis were considered to have immune cell populations downstream of CD3 equal to zero. These patients were included in the analysis of cell frequencies and absolute counts but were excluded from the high-dimensional flow

cytometry analyses. The absolute counts were calculated as the percentage of live lymphocytes from the flow cytometry analysis multiplied by the white blood cell count from the same day for patients where this information was available.

High-dimensional analysis of the flow cytometry data

UMAP learning technique and FlowSOM algorithm creating self-organizing maps (SOMs) (40) were performed using FlowJo Software plugins (FlowJo version 10.7.2 LLC, Ashland, Oregon) and the R programming language. After excluding doublets, dead cells, and CD14/CD19-positive cells, 3000 CD3⁺ events of each patient sample were concatenated (one of the patients had only 2098 events). The UMAP analysis was performed using 15 nearest neighbors with a minimum distance of 0.5. The FlowSOM analysis was performed using 20 meta clusters. The following markers were used for UMAP and FlowSOM analysis for closer phenotyping of the CD3⁺ population: CD4, CD8, CCR7, CD45RO, CD95, CD28, CD127, CD25, CD56, MR1 tetramer, CD1d tetramer, V α 7.2, V δ 1, V δ 2, CD161, CD69, PD1, Tim3, and Lag3. The 20 populations defined by FlowSOM were further identified in each of the individual samples represented by 3000 concatenated events for statistical analysis.

Single-cell RNA sequencing

FACS purification of MAIT and V δ 2 populations from patients was performed using the remaining aliquots (2 to 5 aliquots per patient) of the same sample that was used for the original flow cytometry analysis. Both patient and healthy control samples were stained as described above and sorted using the FACSaria instrument (BD Biosciences). The scRNA-seq of FACS-purified cell suspensions was performed on Chromium instrument (10X Genomics) following the user guide manual for 3' v3.1. Between 6000 and 30,000 cells were targeted for each sample. Samples were multiplexed together on one lane of 10X Chromium following the cell hashing protocol (69). Final libraries were sequenced on the Illumina NovaSeq S4 platform (R1, 28 cycles; i7, 8 cycles; and R2, 90 cycles). The cell-gene count matrix was constructed using the Sequence Quality Control (SEQC) package (70). Viable cells were identified on the basis of library size and complexity, whereas cells with >20% of transcripts derived from mitochondria were excluded from further analysis.

Quality control, demultiplexing, noise reduction, and clustering

The SEQC-generated count matrices for each of the samples were further sanitized and integrated using the shunPykeR pipeline (71). The SEQC-generated count matrices were processed using the scanpy v1.8.2 toolkit (72). Ribosomal genes and genes that were not expressed in any cell were removed from downstream analysis. Each cell was then normalized to a total library size of 10,000 reads, and gene counts were log-transformed using the log(X + 1) formula, in which log denotes the natural logarithm. Principal components analysis (PCA) was applied to reduce noise before data clustering. The knee point (eigenvalues smaller radius of curvature) was used to select the optimal number of PCs to retain for each dataset. PhenoGraph (73) was used to identify clusters within the PCA-reduced data. An adjusted Rand score (range: 5 to 150, step: 5) higher than or equal to 0.8 for four consecutive steps was used to select a robust nearest-neighbor parameter (*k*) for PhenoGraph.

Removal of doublet and contaminant cells

Doublets were predicted using the scrublet (74) python package with default parameters. Although cells were sorted for purified MAIT cell populations after gating out myeloid cells, B cells, and CD3-negative

cells, a small proportion of the cells either expressing the CD68 myeloid marker or not expressing canonical MAIT cell markers (*SLC4A10* and *DPP4*) were detected and marked as contaminating in our scRNA-seq data. The quality of the single cells was computationally assessed on the basis of total counts, number of genes, mitochondrial fraction, and ribosomal fraction per cell, with low total counts, low number of genes (≤ 1000), and high mitochondrial content (≥ 0.2) as negative indicators of cell quality. Cells characterized by more than one negative indicator were considered as poor quality cells. To remove in an unbiased way contaminants and poor quality cells, we assessed them in a cluster basis rather than individually. PhenoGraph-generated clusters with a poor quality profile and/or a high number of contaminating cells (expressing high levels of CD68 or not expressing *SLC4A10* and *DPP4*) were removed on a per-sample basis. In addition, cells marked as doublets were filtered out. After removal of these cells, PCA and unsupervised clustering analysis was reapplied to the filtered data as described in Materials and Methods.

Differential expression analysis

Differential expression analysis between annotated populations of interest was performed using Model-based Analysis of Single-cell Transcriptomics (MAST) (75) with default parameters. Differentially expressed genes were considered statistically significant if the false discovery rate (FDR)-adjusted *P* value was less than 0.05.

Pathway enrichment analysis

The GSEA preranked module from the GSEA v4.1.0 software (52, 53) was used to predict pathway enrichment for (i) MAIT patients versus MAIT healthy and (ii) V δ 2 patients versus V δ 2 healthy comparisons. The coefficient value (log₂-transformed fold change) generated by the differential expression analysis with MAST was used to prerank all differentially expressed genes. Predicted pathways with an FDR ≤ 0.05 were considered as significantly enriched. Human homologs of mouse genes were generated using Ensembl genes 105 (76).

Statistical analysis

P values for α -diversity, taxa abundance, relationship between MAIT/V δ 2 cells and GVHD, metabolic pathway abundances, expression of genes encoding enzymes in these pathways, and analysis of stool metabolites were calculated using a two-sample Wilcoxon rank-sum test using R (version 4.0.3). A multivariable linear regression assessed the association between α -diversity and MAIT cell frequency, both modeled continuously, adjusted for PTCy, conditioning, and age. Overall survival was estimated using Kaplan-Meier methods and compared by MAIT frequency, categorized as above and below the median, using the log-rank test. Nonrelapse mortality (NRM) was estimated using cumulative incidence functions, where relapse was considered a competing risk. NRM was compared by the binary MAIT frequency using Gray's test. *P* values for differences in the flow cytometry populations defined in the FlowSOM analysis and the differences in transcription factor mean fluorescence intensity (MFI) between patients and healthy controls were calculated using a Mann-Whitney test in GraphPad Prism. The FDR approach was used in multiple hypothesis testing to correct for multiple comparisons in the analysis of FlowSOM-defined populations, taxa abundance, and riboflavin/MEP pathway enzymes using an algorithm in R.

SUPPLEMENTARY MATERIALS

www.science.org/doi/10.1126/scitranslmed.abj2829

Figs. S1 to S13

Tables S1 and S2

Data files S1 and S2

Materials Design Analysis Reporting (MDAR) Checklist for Authors

[View/request a protocol for this paper from Bio-protocol.](#)

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MAIT and V#2 unconventional T cells are supported by a diverse intestinal microbiome and correlate with favorable patient outcome after allogeneic HCT

Hana Andrllová, Oriana Miltiadous, Anastasia I. Kousa, Anqi Dai, Susan DeWolf, Sara Violante, Hee-Yon Park, Sudha Janaki-Raman, Rui Gardner, Sary El Daker, John Slingerland, Paul Giardina, Annelie Clurman, Antonio L. C. Gomes, Chi Nguyen, Marina Burgos da Silva, Gabriel K. Armijo, Nicole Lee, Roberta Zappasodi, Ronan Chaligne, Ignas Masilionis, Emily Fontana, Doris Ponce, Christina Cho, Amy Bush, Lauren Hill, Nelson Chao, Anthony D. Sung, Sergio Giralte, Esther H. Vidal, Kinga K. Hosszu, Sean M. Devlin, Jonathan U. Peled, Justin R. Cross, Miguel-Angel Perales, Dale I. Godfrey, Marcel R. M. van den Brink, and Kate A. Markey

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Unconventional transplant support

Numerous studies have suggested that a diverse intestinal microbiome is linked to better outcomes in recipients of allogeneic hematopoietic cell transplantation (allo-HCT). Andrllová *et al.* look to understand how unconventional T cell subsets facilitate this outcome in a cohort of 174 patients with allo-HCT. They observed that a diverse intestinal microbiome at early times after allo-HCT is associated with higher numbers of innate-like mucosal-associated invariant T (MAIT) cells, and this correlated with a higher prevalence of the V#2 subpopulation of $\alpha\alpha$ T cells and a reduced incidence of acute graft-versus-host disease (aGVHD). Further analysis indicated that a diverse intestinal microbiome could produce ligands that promote maintenance of activated and functional MAIT and V#2 cells following allo-HCT. These findings provide evidence of the immunological link between a diverse microbiome and maintenance of unconventional T cell subsets.

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